DIOGO GOMES DE ARAÚJO

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Education

Master's **Sep 2025 – Present**
Computer Science *Universidade do Porto (FCUP)*
Focused on Distributed Systems, Programming Languages and Formal Verification.

Bachelor's **Sep 2022 – Jul 2025**
Computer Science and Engineering *Universidade da Beira Interior*
Final project presented at Inforum 2025 in Évora.

Research and Projects

From Ideal Clocks to Buildable Ones | Applied Cryptography, Model Checking, Blockchain Protocols

- Currently starting research into using **model checking** techniques to compare realistic clock constructions with ideal ones used in blockchain bridges specified in the Universal Composability (UC) framework in collaboration with Input Output Global.

Threshold Signature Library and Wallet for Nano | Rust, Dalek (Ed25519), Tokio, Dioxus (WASM)

- Implemented a library for **FROST threshold signing** on Nano, which is not natively supported by the protocol.
- Built a desktop application for creating shared accounts and co-signing transactions remotely, hiding the peer-to-peer signing protocol behind a simple GUI.

PoW Blockchain from Decentralized Auctions | Rust, LibP2P, Dalek (Ed25519), Tokio, Docker, Ratatui (CLI)

- Built a network overlay for peer discovery on top of **Kademlia** (Ethereum's network protocol).
- Designed a **Proof-of-Work blockchain** using BLAKE2b hashing, Ed25519 signatures, Merkle trees and Longest Chain Consensus.
- Created an auction client that uses the chain as a public ledger for recording and verifying bids.
- Simulated bootstrap, honest and malicious nodes in Docker to study adversarial behaviour.

Work Experience

INESC TEC **Jan 2026 – Apr 2026**
Researcher *Hybrid*

- Maintained **IoT gateway microservices** for communication with Distributed Energy Resources (DER).
- Deployed a **Zero Trust** setup (Teleport) giving secure remote access to every IoT device in the Power and Energy Systems lab.

STAR Junior Enterprise **May 2024 – Aug 2025**
Project Manager and Backend Developer *Remote*

- Developed **backend REST APIs** for mobile and web applications.
- Led project teams and worked directly with clients to deliver on requirements and on time.
- Screened candidates during recruitment.

Certifications

2nd Place in IEEE RetroJam 2025 | IEEE UP Student Branch **Oct 2025**
Built a complete game in 48 hours with Rust and Raylib, including game mechanics, rendering and input handling.

Participation in Inforum 2025 | Inforum **Sep 2025**
Presented my Bachelor's final project to researchers, professionals and students.

Industrial Robotics Fundamentals | Universidade da Beira Interior **Jul 2025**

Volunteering | Missão País **Feb 2024**

Participation in RoboCup 2016 | RoboCup Federation **Jun 2016**

1st Place in the National Robotics Championship | Instituto Politécnico de Bragança **May 2016**

Technical Skills

Languages: Rust, OCaml, Haskell, C, Java, SQL
Distributed Systems & Networking: libp2p, QUIC, gRPC
Systems & Concurrency: UNIX, POSIX, OpenMP, STM, Tokio
Tools & Infrastructure: Git, Docker, Teleport, PostgreSQL, Raylib
Formal Methods & Testing: Quint, Rocq, QuickCheck
Spoken Languages: Portuguese (Native), English (B2 First, Cambridge)